

# PE-EC701A — Microwave Theory and Technique

Semester VII • 3L:0T:0P • 3 Credits

## Module 7 Exam Notes: Microwave Antennas

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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## 1. Antenna Parameters

Antennas are transitions between guided waves and free-space radiation. They are characterized by several fundamental parameters.

**Effective Aperture ( $A_e$ ):** A parameter describing the power-collecting ability of an antenna. It is the ratio of received power to the incident wave power density:

$$A_e = \frac{P_{\text{rec}}}{W_i} \quad (\text{m}^2) \quad (1)$$

### Core Antenna Parameters

- 1. Directive Gain ( $G_d$ ) & Directivity ( $D$ ):** Directivity is the maximum value of directive gain in the peak radiation direction.

$$D = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}} \quad (2)$$

- 2. Aperture Gain Relation:**

$$G = D \cdot \eta_{\text{rad}} = \frac{4\pi A_e}{\lambda^2} \quad (3)$$

- 3. Beamwidth (HPBW & FNBW):** Half-Power Beam Width (HPBW) is the angular span between the 3 dB down points. First-Null Beam Width (FNBW) is the span between the first zero-intensity nulls.

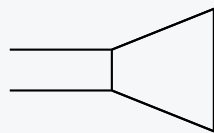
$$\text{FNBW} \approx 2 \times \text{HPBW} \quad (4)$$

## 2. Ground-Based Systems

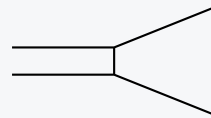
### 2.1 Horn Antenna

Horn antennas flare the end of a waveguide to match the waveguide impedance to the intrinsic impedance of free space ( $377 \Omega$ ), minimizing reflection.

#### Horn Antenna Types



Pyramidal Horn



Sectoral Horn

#### Pyramidal Horn Gain:

$$G = \frac{4\pi}{\lambda^2} \epsilon_{ap} A = \frac{4\pi}{\lambda^2} \epsilon_{ap} (a_e b_e) \quad (5)$$

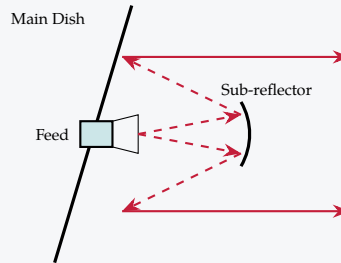
where  $a_e, b_e$  are the aperture widths and  $\epsilon_{ap}$  is the aperture efficiency.

## 2.2 Parabolic Reflector Antenna

Reflector antennas focus a feed horn's spherical wavefronts into highly directive plane waves.

**Cassegrain Feed:** Uses a primary parabolic reflector, a secondary hyperbolic sub-reflector at the focus, and a feed horn placed at the vertex of the primary dish.

### Cassegrain Feed Ray-Tracing Geometry



**Gain:**  $G = \left(\frac{\pi D}{\lambda}\right)^2 \eta_{ap}$  where  $D$  is the diameter of the dish.

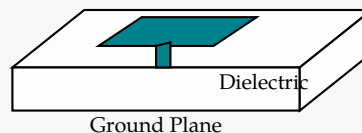
## 3. Airborne and Planar Antennas

Airborne antennas must be aerodynamic, low-weight, and often conformal (matching the contour of the aircraft skin).

### 3.1 Microstrip Patch Antenna

A microstrip patch antenna consists of a radiating metallic patch printed on one side of a thin dielectric substrate with a continuous ground plane on the other.

### Microstrip Patch Antenna Structure



#### Feed Techniques:

1. **Microstrip Line Feed:** Easy to print, easy to match using inset cuts.
2. **Coaxial Probe Feed:** Feed probe is connected through a hole from the bottom ground, matching is done by adjusting the insertion point.
3. **Aperture-Coupled Feed:** Non-contact feed where a slot in the ground plane couples energy from a bottom line.
4. **Proximity-Coupled Feed:** Multi-layer line printed directly below the patch.

## Quick Revision — Module VII

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### Key Formulas at a Glance

|                     |                                                                     |
|---------------------|---------------------------------------------------------------------|
| Effective Aperture  | $A_e = \frac{\lambda^2}{4\pi} G$                                    |
| Horn Antenna Gain   | $G = \frac{4\pi}{\lambda^2} \epsilon_{ap} (a_e b_e)$                |
| Parabolic Dish Gain | $G = \left(\frac{\pi D}{\lambda}\right)^2 \eta_{ap}$                |
| Parabolic HPBW      | $\text{HPBW} \approx 70 \frac{\lambda}{D} \quad (\text{degrees})$   |
| Patch Length        | $L \approx \frac{c}{2f_0 \sqrt{\epsilon_{\text{eff}}}} - 2\Delta L$ |

### One-Line Definitions

- **Directivity:** The ratio of maximum radiation intensity to average radiation intensity.
- **Pyramidal Horn:** A waveguide flaring in both electric and magnetic field planes to match free space.
- **Cassegrain Feed:** A dual-reflector antenna utilizing a parabolic dish and a hyperbolic sub-reflector.
- **Aperture Efficiency:** The ratio of the effective antenna aperture to its physical geometric area.
- **Polarization:** The orientation of the electric field vector of the radiated wave as it propagates.
- **Conformal Antenna:** An antenna designed to follow the curved contour of an host aircraft or vehicle.
- **Aperture Coupling:** A non-contact microstrip feeding technique using a slot in a shared ground plane.

### Frequently Asked / Viva Questions

**Q1. State the relation between antenna gain, directivity, and efficiency.**

$G = \eta_{\text{rad}} \cdot D$ , where  $\eta_{\text{rad}}$  is the radiation efficiency. For a lossless antenna, gain equals directivity ( $G = D$ ).

**Q2. What are the advantages of Cassegrain feeds over focal-point feeds?**

Shorter waveguide feed lines (reducing insertion loss), lower spillover noise from the ground, and easier access to the feed horn at the vertex.

**Q3. Explain the physical operation of a microstrip patch antenna.**

The patch acts as a resonant cavity. Radiation occurs primarily from the fringing fields at the open-circuited edges of the patch.

**Frequently Asked / Viva Questions (contd.)****Q4. Why does flaring a waveguide into a horn antenna increase directivity?**

Flaring increases the physical aperture area of the waveguide. Since  $D \propto A/\lambda^2$ , a larger aperture yields higher directivity and narrower beamwidth.

**Q5. What is circular polarization and when is it preferred?**

A wave polarization where the E-field vector rotates in a circle. It is preferred in satellite links because it is immune to Faraday rotation in the ionosphere.

**Q6. Why do planar antennas require low-profile feed techniques in airborne systems?**

To maintain aerodynamic properties, preventing wind drag, turbulence, and structural damage to the aircraft.

**Q7. State the difference between E-plane and H-plane sectoral horns.**

An E-plane horn flares only in the direction of the electric field. An H-plane horn flares only in the direction of the magnetic field.

**Q8. How does the dielectric substrate thickness affect microstrip patch performance?**

Thicker substrates increase bandwidth and radiation efficiency but lead to surface-wave excitation, causing radiation pattern degradation.

**Q9. What is FNBW and how is it related to HPBW?**

FNBW is First-Null Beamwidth. For symmetric main lobes, FNBW is approximately twice the HPBW ( $\text{FNBW} \approx 2 \times \text{HPBW}$ ).

**Q10. What is the purpose of the sub-reflector in a Cassegrain antenna?**

To reflect the feed signals onto the main dish, allowing the feed electronics to be placed behind the main dish rather than suspended at the focus.

**Q11. How do lens antennas work at microwave frequencies?**

They use dielectric materials to collimate spherical wavefronts into plane waves, similar to optical lenses.

**Q12. What is the typical gain range of a standard horn antenna?**

Typically between 10 dB and 25 dB, depending on flaring and operating frequency.

## Comparison of Microwave Antennas

| Antenna Type            | Typical Gain            | Bandwidth              | Physical Profile              |
|-------------------------|-------------------------|------------------------|-------------------------------|
| <b>Horn Antenna</b>     | Moderate (12 – 25 dBi)  | Wide (waveguide range) | Bulky flared structure        |
| <b>Parabolic Dish</b>   | Very High (30 – 50 dBi) | Wide                   | Very bulky reflector assembly |
| <b>Microstrip Patch</b> | Low (5 – 8 dBi)         | Very Narrow (2% – 5%)  | Flat, lightweight, conformal  |
| <b>Lens Antenna</b>     | High (20 – 30 dBi)      | Moderate               | Heavy dielectric component    |